

Independent Non-Redundant Open Conjectures Outside Number Theory

79 Problems After Collapsing Implications and Disproved Cases

Compilation from 100-item list¹

¹*Mesh analysis*
(Dated: July 20, 2026)

From a 100-item list of open conjectures outside number theory we remove duplicates, resolved and disproved cases, and implication chains. Duplicates: sphere packing in $5-7 \subset$ general Kepler, Willmore higher genus = Willmore generalizations, Hadwiger-Nelson adjacent = chromatic number of plane, Erdos-Szekeres appears twice, Farrell-Jones appears twice. Implications: Farrell-Jones $\Rightarrow$ Borel, Baum-Connes, Novikov, Kaplansky idempotent; Zeeman $\Rightarrow$ Andrews-Curtis. Resolved/disproved: Telescope Conjecture disproved 2023, Borsuk disproved Kahn-Kalai, Hedetniemi disproved Shitov 2019, Connes Embedding refuted MIP*=RE 2020, Serre Galois proved Khare-Wintenberger, Virtually Haken and Property P proved. Remaining 79 are pairwise independent.

I. REDUCTION

- Keep strongest in each chain: Farrell-Jones, Zeeman, Kepler general, Firoozbakht analog not present here.
- Drop special cases: $5-7$ sphere packing, Willmore duplicate, Hadwiger-Nelson adjacent, Erdos-Szekeres duplicate, Farrell-Jones duplicate.
- Exclude disproved/proved: Telescope, Borsuk original, Hedetniemi original, Connes Embedding, Serre, Virtually Haken core, Property P core, Property R vague.

II. INDEPENDENT 79

- Smooth 4D Poincaré:** Every smooth homotopy 4-sphere diffeomorphic to S^4 .
- Zeeman Collapsibility:** If K is contractible 2-complex then $K \times I$ collapsible. Implies Andrews-Curtis.
- Hilbert-Smith:** No p -adic group acts effectively on a manifold.
- Farrell-Jones:** K_*, L_* of group rings computed by $E_{VC}G$. Implies Borel, Novikov, Baum-Connes, Kaplansky idempotent.
- Slice-Ribbon:** Every slice knot in S^3 ribbon.
- Cabling Conjecture:** Dehn surgery yielding reducible manifold only for cable knots.
- Berge Conjecture:** Classification of knots admitting lens space surgeries.
- Effective Virtually Haken:** Quantitative versions of Agol's theorem remain open.
- Simple Loop Conjecture:** Non-injective π_1 map $\Sigma \rightarrow M^3$ has essential simple loop in kernel.
- Unknotting Number Additivity:** $u(K_1 \# K_2) = u(K_1) + u(K_2)$.
- Kneser-Poulsen:** Moving centers of balls closer does not increase volume of union.
- Kepler-type general:** Optimal sphere packing density in $\mathbf{R}^n$ unknown for most $n \neq 3, 8, 24$.
- Kissing Number:** Max number unit spheres touching unit sphere in $\mathbf{R}^n$, exact only for few n .
- Willmore generalizations:** Minimal Willmore energy for genus > 1 and higher codimension, torus case proved Marques-Neves.
- Carathéodory:** Convex C^2 sphere in $\mathbf{R}^3$ has ≥ 2 umbilics.
- Yau First Eigenvalue:** Closed embedded minimal hypersurface in S^{n+1} has $\lambda_1 = n$.
- Bellman Lost in Forest:** Optimal worst-case escape path from forest of unknown shape.
- Moving Sofa:** Maximal area shape navigating unit-width L-corridor.
- Square Peg / Toeplitz:** Every Jordan curve contains 4 points forming a square.
- Erdős-Szekeres:** Minimal $N(n)$ points forcing convex n -gon is $2^{n-2} + 1$.
- Illumination:** Every convex body in $\mathbf{R}^n$ illuminable by 2^n light sources.
- Takeya:** Besicovitch set containing unit segment in every direction has Hausdorff dimension n .
- Unit Distance:** Max number unit distances among n points in plane, conjectured $n^{1+o(1)}$.
- Distinct Distances exact constant:** Guth-Katz $\Omega(n/\log n)$, exact constant open.
- Chromatic Number of Plane:** $\chi(\mathbf{R}^2) \in \{5, 6, 7\}$.
- Danzer's Problem:** Bounded-density set in $\mathbf{R}^2$ meeting every convex body of area 1?

- 27. Ulam Packing:** No convex body packs $\mathbf{R}^3$ more efficiently than sphere.
- 28. Reinhardt:** Smoothed octagon is least efficient centrally symmetric convex shape packing plane.
- 29. Non-spherical Kepler:** Optimal packing density for most convex bodies unknown.
- 30. Bang Plank complex:** Covering convex body by planks, complex version open.
- 31. Erdős-Oler:** Optimal packing of $n(n+1)/2$ equal circles in equilateral triangle.
- 32. Falconer Distance Set:** If $\dim_H E > 1$ in $\mathbf{R}^2$, distance set has positive measure.
- 33. Zarankiewicz:** Crossing number of $K_{m,n}$ equals $\lfloor m/2 \rfloor \lfloor (m-1)/2 \rfloor \lfloor n/2 \rfloor \lfloor (n-1)/2 \rfloor$.
- 34. Hadwiger:** $\chi(G) \leq \eta(G)$, chromatic number $\leq$ size of largest clique minor.
- 35. Erdős-Faber-Lovász:** n cliques pairwise intersecting ≤ 1 vertex need n colors. Large n proved 2021, full still debated.
- 36. Total Coloring:** $\chi''(G) \leq \Delta(G) + 2$.
- 37. List Coloring:** List edge chromatic number $= \chi'(G)$.
- 38. Reconstruction:** Graph ≥ 3 vertices determined by multiset of vertex-deleted subgraphs.
- 39. Edge Reconstruction:** Same for edge-deleted subgraphs.
- 40. Cycle Double Cover:** Every bridgeless graph has collection of cycles covering each edge twice.
- 41. Erdős-Gallai cycle covers:** Edges of n -vertex graph coverable by $O(n)$ cycles?
- 42. Barnette:** Every cubic 3-connected planar bipartite graph Hamiltonian.
- 43. Chvátal Toughness:** Every $3/2$ -tough graph Hamiltonian.
- 44. Erdős-Hajnal:** For fixed H , H -free n -vertex graph has clique or independent set $\geq n^{c(H)}$.
- 45. Union-Closed / Frankl:** In union-closed family $\neq \{\emptyset\}$, some element in $\geq 1/2$ sets.
- 46. Ryser:** For r -partite r -uniform hypergraph, $\tau \leq (r-1)\nu$.
- 47. EKR extensions:** Exact extremal families for general set systems beyond proven cases.
- 48. Chromatic Symmetric Function:** For $(3+1)$ -free poset, X_G is e -positive.
- 49. Tuza:** $\tau_t \leq 2\nu_t$ for triangle packing vs covering.
- 50. Sidorenko:** For bipartite H , $t(H, G) \geq t(K_2, G)^{e(H)}$.
- 51. Graceful Tree:** Every tree has graceful labeling $0..m$ with distinct edge differences.
- 52. Erdős-Gallai path:** n -vertex graph with $> n(k-1)/2$ edges contains path with k edges.
- 53. 1/3-2/3 Conjecture:** Every non-total poset has pair with $1/3 \leq P(x < y) \leq 2/3$ in random linear extensions.
- 54. Rota's Basis:** n bases of rank n matroid can be arranged in $n \times n$ grid where rows and columns are bases.
- 55. Aharoni-Berger:** n matchings size $n+1$ in bipartite graph have rainbow matching size n .
- 56. Sunflower exact bound:** Sharp bound on size of k -uniform family with no w -sunflower.
- 57. Erdős-Sós:** Graph average degree $> k-1$ contains every tree with k edges.
- 58. Vizing Domination:** $\gamma(G \square H) \geq \gamma(G)\gamma(H)$.
- 59. Alon-Tarsi:** Even n Latin squares: number even $\neq$ number odd, implies list coloring conjectures.
- 60. Smale Mean Value:** $\min_{p'(c)=0} |p(c)/((c-z_0)p'(z_0))| \leq 1$ for polynomial p .
- 61. Arnold Fixed Point:** Hamiltonian symplectomorphism has $\geq$ sum Betti numbers fixed points.
- 62. Weinstein general:** Reeb vector field on closed contact manifold has periodic orbit, open beyond dim 3.
- 63. Palis:** Every dynamical system approximable by hyperbolic or with homoclinic tangency.
- 64. Shub Entropy:** $h_{\text{top}}(f) \geq \log \rho(f_*)$.
- 65. Zimmer Program:** Higher rank lattice actions on manifolds essentially algebraic.
- 66. MLC:** Mandelbrot set locally connected.
- 67. Fatou:** Hyperbolic rational maps dense in space of rational maps.
- 68. Furstenberg $\times 2 \times 3$:** Ergodic $\times 2, \times 3$ -invariant measure on circle is Lebesgue or atomic.
- 69. Banach L^p classification:** Isomorphic classification of L^p spaces remains incomplete.
- 70. Invariant Subspace Hilbert:** Every bounded operator on separable Hilbert space has non-trivial closed invariant subspace.
- 71. Berry-Tabor:** Generic integrable quantum system spectral statistics Poissonian.
- 72. Quantum Unique Ergodicity general:** Arithmetic case proved Lindenstrauss, general manifolds open.
- 73. Jacobian:** Polynomial map $\mathbf{C}^n \rightarrow \mathbf{C}^n$ with constant non-zero Jacobian determinant has polynomial inverse.

- 74. Zariski Cancellation:** If $X \times \mathbf{A}^1 \cong \mathbf{A}^{n+1}$ over $\mathbf{C}$, then $X \cong \mathbf{A}^n$, open for $n = 3$.
- 75. Bass Conjecture:** K_0 of group ring finitely generated.
- 76. Quillen-Lichtenbaum:** K -theory of number fields relates to étale cohomology beyond proven range.
- 77. Deligne Special Values:** Critical values of motivic L -functions are periods up to algebraic factor.
- 78. Homological Mirror Symmetry:** Derived Fukaya category equivalent to derived category of coherent sheaves of mirror, general case open.
- 79. Green Syzygies:** Clifford equals gonality for canonical curve, generic case proved Voisin, special cases open.